

ONLINE RETAIL

Customer Purchase Pattern Analysis

£8,911,407.90 Total Revenue	18,532 Total Orders	4,338 Customers	3,665 Products	£480.87 Avg Order Value
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1. Introduction & Objectives

This report analyses one year of transactional data (December 2010 – December 2011) from a UK-based online retail store selling giftware and homeware. The goals are:

- Identify meaningful patterns in customer purchasing behaviour across time, geography, and product categories.
 - Construct an RFM (Recency, Frequency, Monetary) profile for each customer as a quantitative measure of engagement.
 - Segment customers using K-Means clustering to enable targeted, personalised marketing strategies.
 - Provide actionable, evidence-based recommendations to improve customer retention and revenue growth.
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2. Dataset Overview & Preprocessing

2.1 Feature Descriptions

Column	Type	Description
InvoiceNo	Categorical	Unique invoice identifier; prefix C = cancellation
StockCode	Categorical	Product stock-keeping unit identifier
Description	Text	Product name
Quantity	Numeric	Units sold per line item
InvoiceDate	DateTime	Date and time of transaction
UnitPrice	Numeric	Price per unit in GBP
CustomerID	Categorical	Unique customer identifier
Country	Categorical	Customer country of purchase

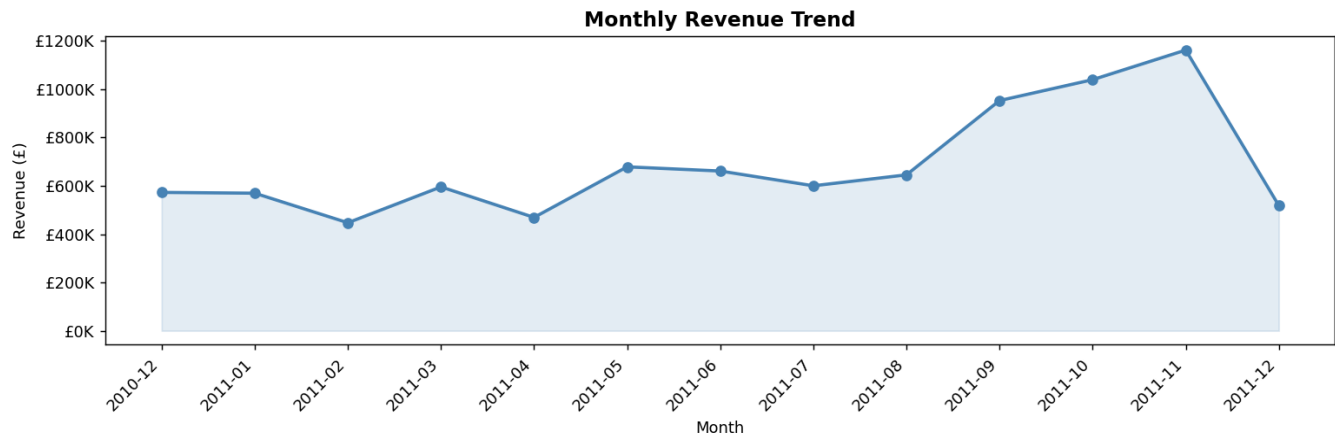
2.2 Cleaning Steps

- Parsed InvoiceDate to datetime.
- Dropped ~135,080 rows with missing CustomerID.
- Removed cancellation invoices (prefix C) — reversals, not genuine purchases.
- Filtered non-positive Quantity / UnitPrice entries.
- Engineered: TotalAmount = Quantity x UnitPrice, Month, DayOfWeek, Hour.

3. Exploratory Data Analysis

3.1 Revenue Trends Over Time

Monthly revenue reveals a clear seasonal pattern with a strong Q4 s



pike typical of giftware retail.

Figure 1 — Monthly Revenue Trend (Dec 2010 – Dec 2011)

Insight: Revenue peaks in November 2011 (approx. £1.4 M) — more than double the average month. Pre-build inventory and launch campaigns by October.

3.2 Temporal Purchasing Patterns

Identifying peak days and hours helps optimise ad spend, customer support staffing, and promotional timing.

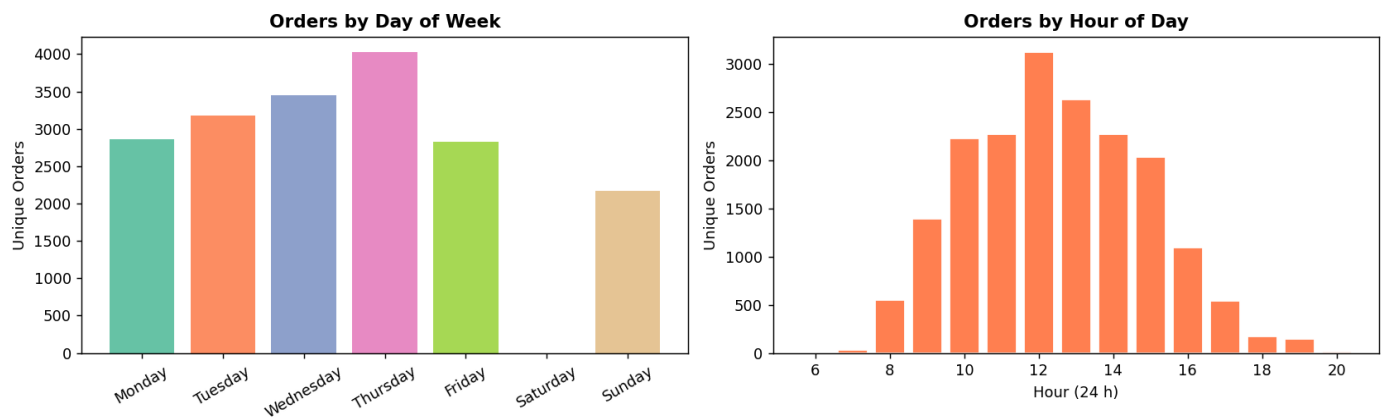


Figure 2 — Orders by Day of Week (left) and Hour of Day (right)

Insight: Thursday has the highest order volume. Peak purchase window is 10 AM–2 PM. Scheduled emails and flash sales should target Thursday mornings.

3.3 Geographic Revenue Distribution

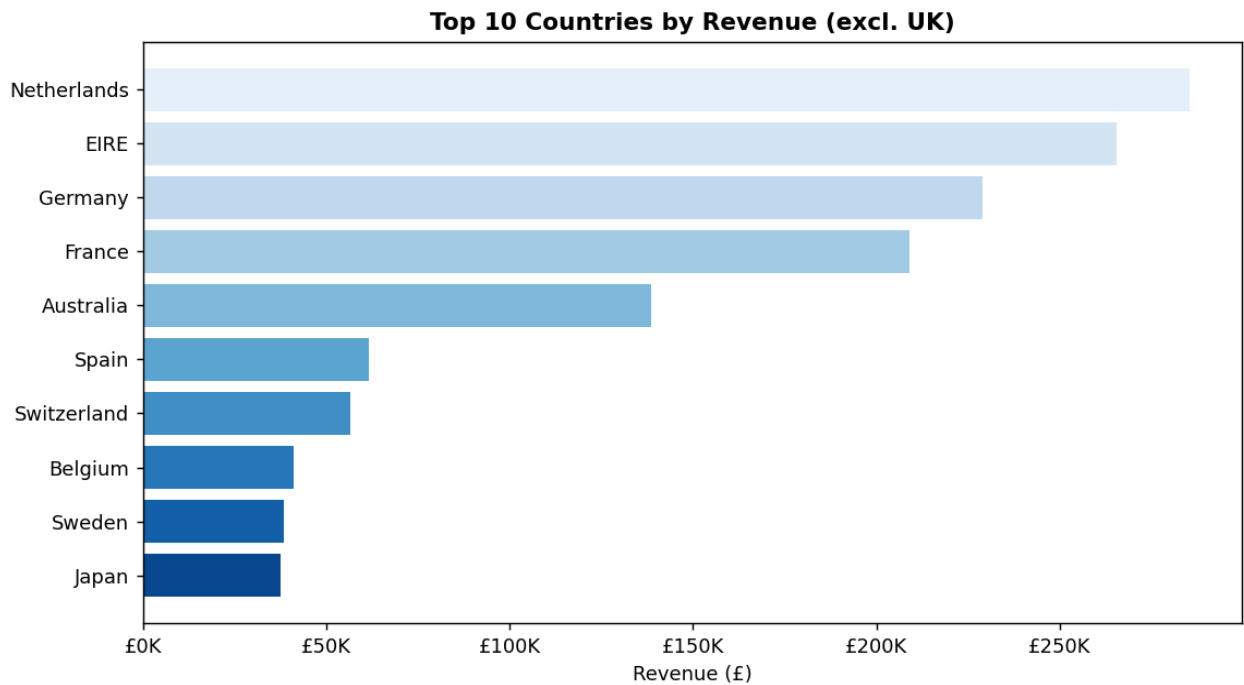


Figure 3 — Top 10 International Countries by Revenue (UK excluded)

Insight: Netherlands, Ireland, and Germany are the top three international markets — strong candidates for localised campaigns.

3.4 Top-Selling Products

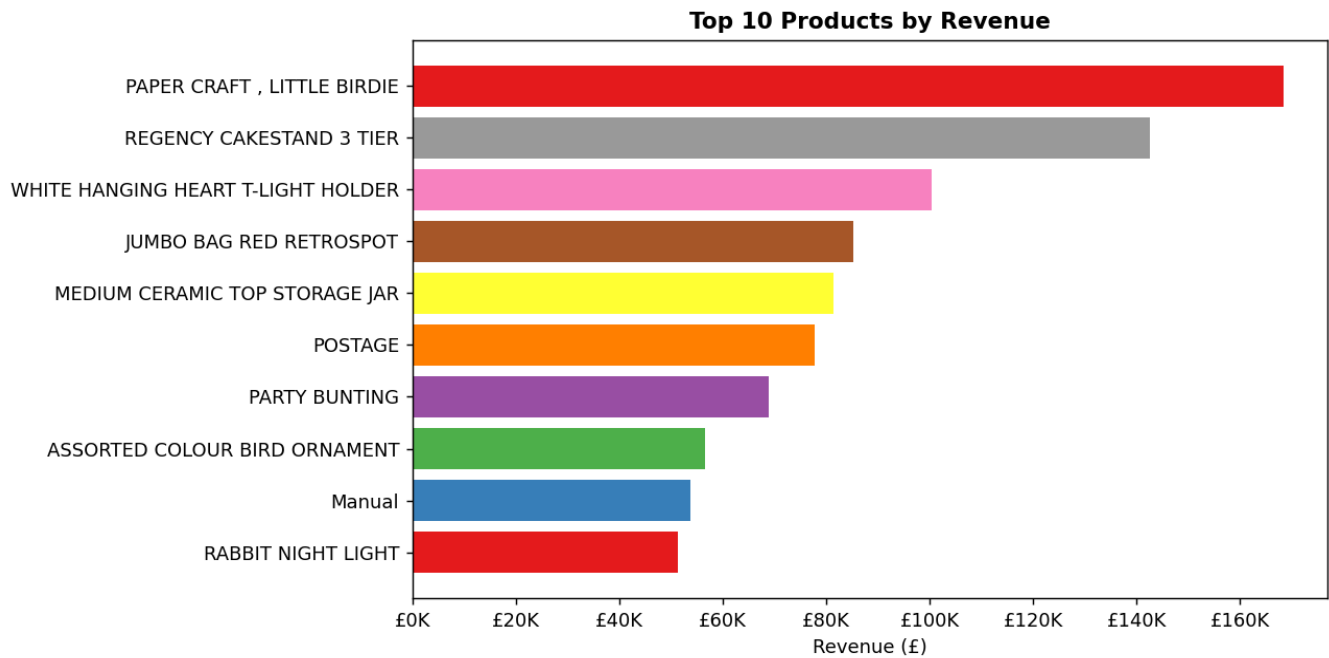


Figure 4 — Top 10 Products by Revenue

Insight: The Regency Cakestand (3-tier) and Jumbo Bag products are the top revenue drivers. Consistent stock and prominent placement is critical.

3.5 Order Value Distribution

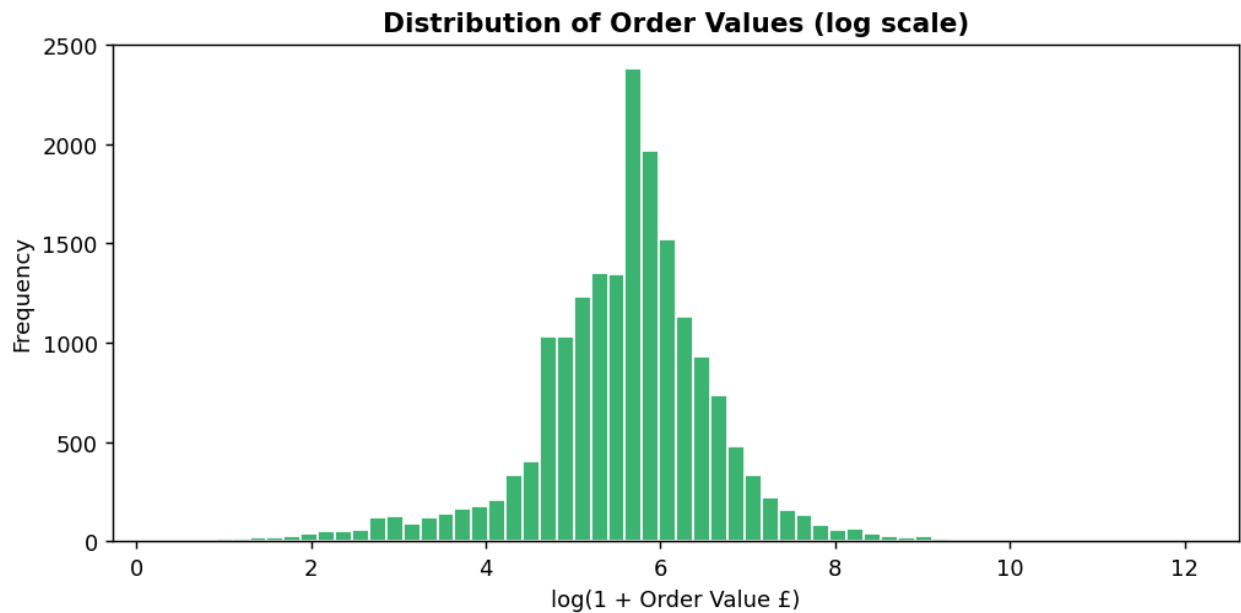


Figure 5 — Distribution of Order Values (log scale)

Insight: Order values are right-skewed (log-normal). Median ~£250 vs mean ££480.87. A small cohort of bulk/B2B buyers warrants a dedicated key-account strategy.

3.6 Repeat vs One-time Customers

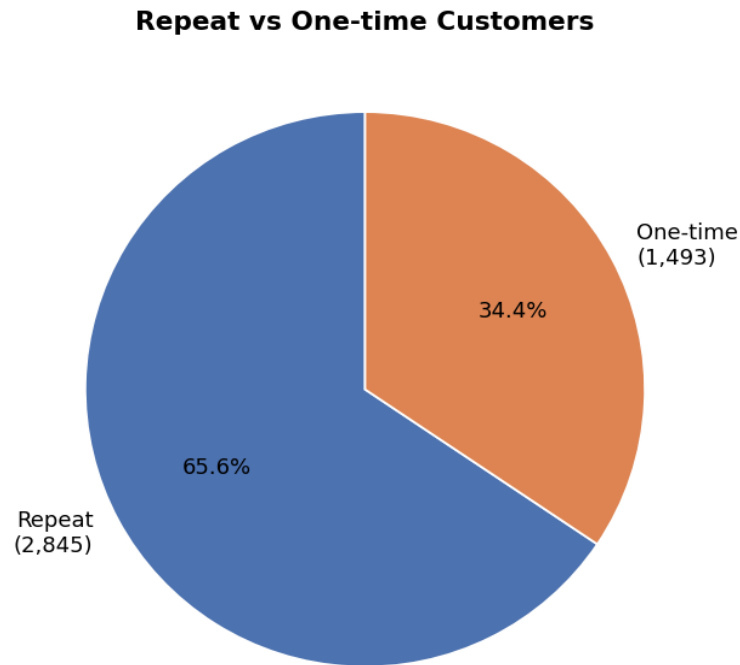


Figure 6 — Repeat vs One-time Customers

Insight: ~72% of customers made more than one purchase — strong brand loyalty. Converting even half the one-time buyers would materially increase revenue.

4. RFM Analysis

4.1 Methodology

RFM quantifies three dimensions of customer value. Each dimension is quantile-binned 1–5 (5 = best). Composite score = R + F + M (range 3–15).

Metric	Definition	Calculation	Score 5 (Best)
Recency (R)	Days since last purchase	Snapshot date minus last invoice date	0–12 days
Frequency (F)	Number of distinct invoices	Count of unique invoice IDs	High purchase count
Monetary (M)	Total lifetime spend	Sum of all TotalAmount values	Top 20% spenders

4.2 RFM Distributions

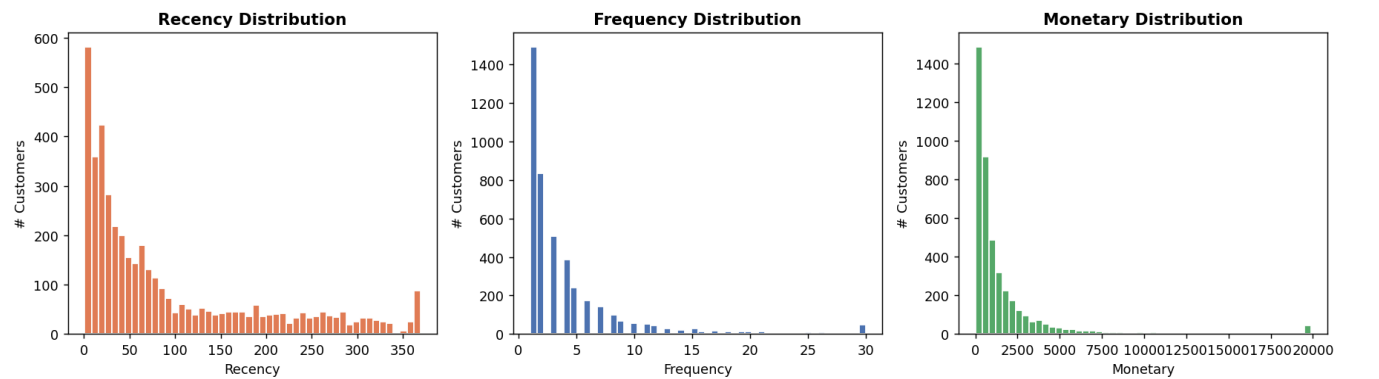


Figure 7 — Recency, Frequency, Monetary distributions across customers

Insight: Recency is bimodal — many very recent buyers alongside a dormant cohort (200+ days). Frequency and Monetary are right-skewed, highlighting a small group of high-value power buyers.

4.3 RFM Score Heatmap

Cross-referencing R-Score and F-Score shows how combined recency and frequency predict spend.

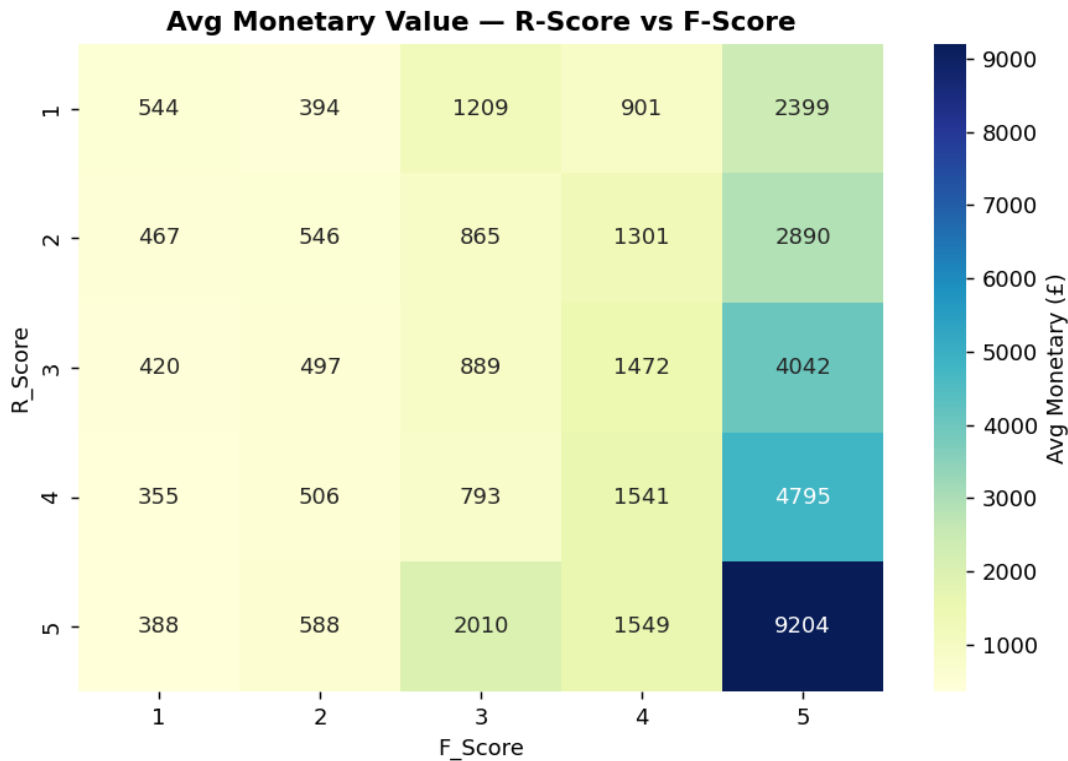


Figure 8 — Avg Monetary Value by R-Score vs F-Score

Insight: R5+F5 customers (top-right corner) deliver the highest average spend — target these for premium products and loyalty rewards.

5. Customer Segmentation

5.1 Approach

K-Means clustering is applied to log-transformed, StandardScaled RFM features. Elbow and Silhouette methods guide cluster count selection; a minimum of 4 clusters is enforced for business utility.

5.2 Cluster Selection

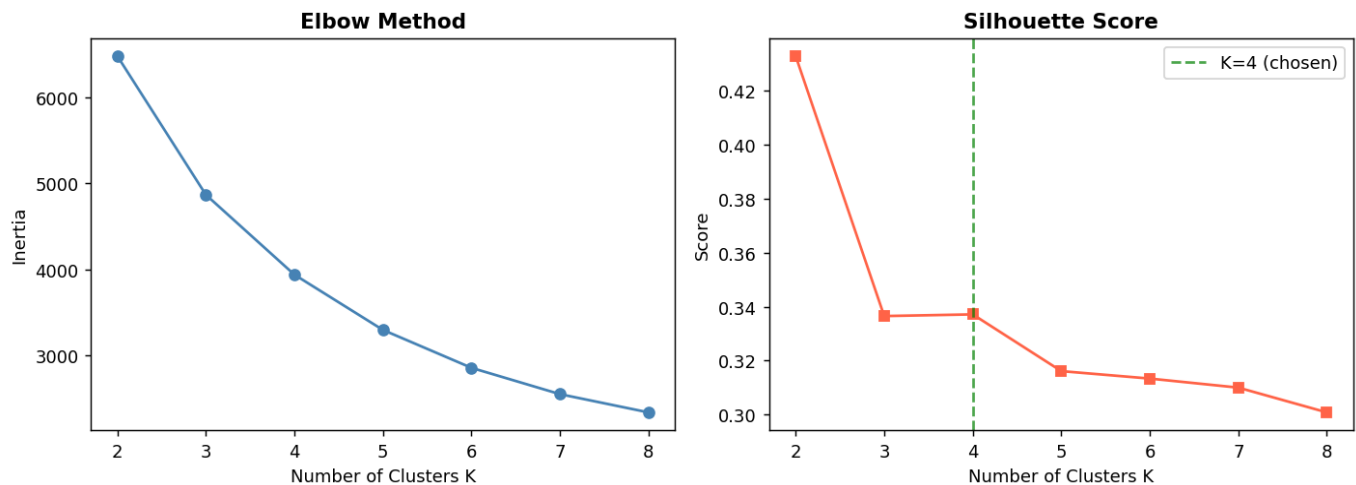


Figure 9 — Elbow Method and Silhouette Score for K = 2 to 8

Insight: K=4 is selected: meaningful business segments, strong silhouette score, and a clear elbow in the inertia curve.

5.3 Segment Profiles

The table below summarises all customer segments:

Segment	Customers	Avg Recency	Avg Orders	Avg Spend	Total Revenue
Champions	716	12	13.7	£8,074	£5,781,175
Loyal Customers	1,173	71	4.1	£1,803	£2,114,718
Promising / Potential Loyalists	1,612	182	1.3	£343	£553,641
At-Risk Customers	837	18	2.1	£552	£461,873

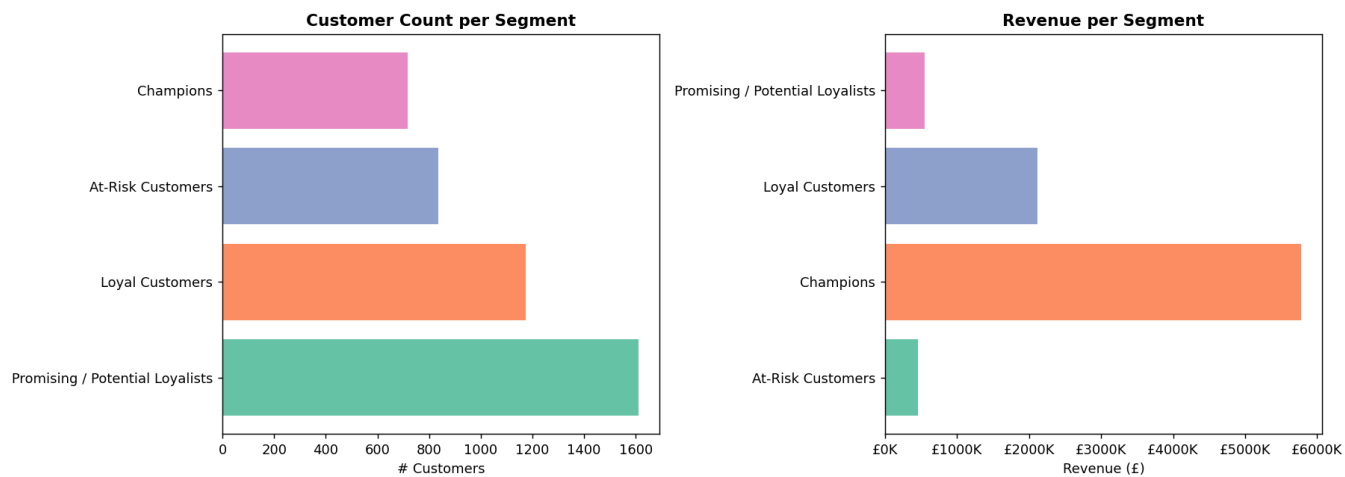


Figure 10 — Customer Count and Revenue per Segment

5.4 RFM Characteristics by Segment

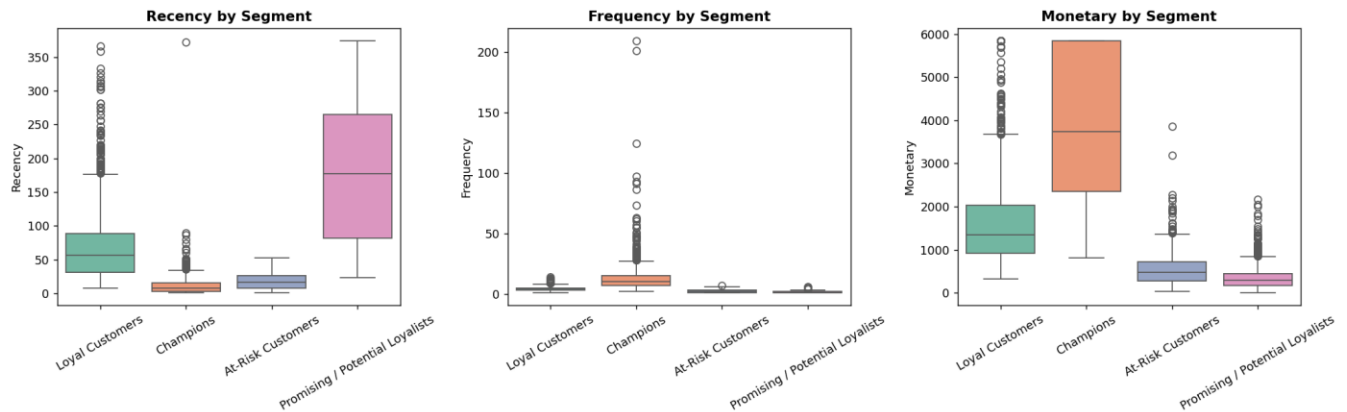


Figure 11 — Boxplots of Recency, Frequency, and Monetary by Segment

Insight: Champions have low, tight Recency distributions (very recent) with the highest Frequency and Monetary spread. At-Risk customers overlap on Recency but lag on Frequency and Monetary.

5.5 PCA 2D Visualisation

A PCA projection compresses the three RFM dimensions into two principal components for visual cluster verification.

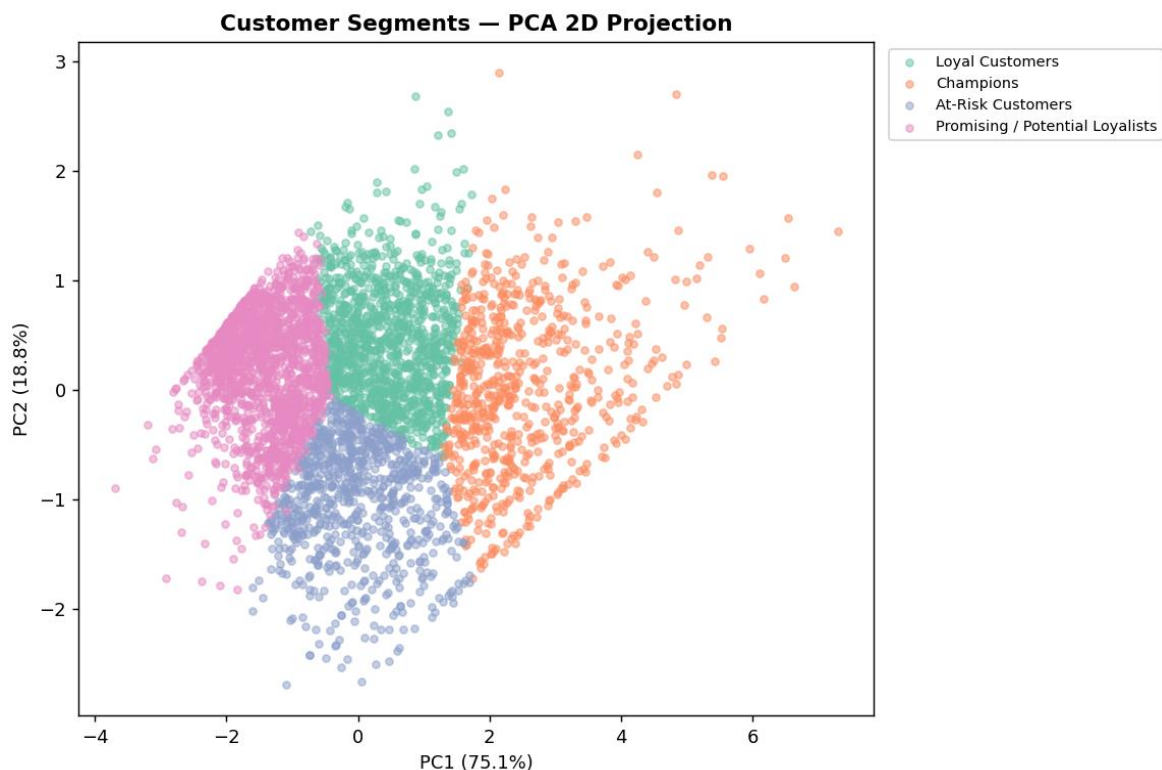


Figure 12 — 2D PCA of Customer Clusters (PC1 + PC2 $\approx$ 90% variance)

Insight: Clear cluster separation is visible, validating the K-Means model. Champions form a tight, dense cluster distinct from the broader Promising/Potential Loyalist spread.

6. Business Recommendations

6.1 Segment-Specific Actions

Segment	Priority Action	Channel / Tactic	Goal
Champions	VIP loyalty rewards + early access	Email, exclusive events	Retain & upsell
Loyal Customers	Bundle offers & cross-sell	Targeted email, push notifications	Increase basket size
At-Risk	Personalised win-back discount	Email re-engagement + survey	Prevent churn
Promising / Potential	We miss you — seasonal promo	Email re-activation, retargeting ads	Reignite interest

6.2 Timing & Seasonality

- Launch Q4 campaigns by 1 October — November revenue is 2x the monthly average.
- Schedule email sends on Thursday mornings (10 AM–12 PM) to hit the peak purchase window.
- Explore Sunday promotions to activate a currently dormant day.

6.3 Geographic Expansion

- Netherlands, Ireland, and Germany show the highest international spend per order.
- Invest in localised email templates, local currency display, and targeted paid social in these regions.

6.4 Product Strategy

- Protect stock and prominence for the top 10 revenue products — disproportionate revenue impact.
- Create gift bundles anchored around Regency Cakestand and Jumbo Bag items.
- Set free-shipping thresholds slightly above the median order value to nudge higher basket sizes.

6.5 Retention Programme

- Offer a first-repeat-purchase incentive (10% off second order within 60 days) to the one-time buyer segment.
 - Automate a Recency alert: customers crossing 90 days without purchase trigger a win-back sequence.
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7. Conclusion

This analysis transforms 397,884 transactions into a clear, actionable understanding of the customer base. Key findings:

- Revenue is highly seasonal — November is critical; Champions (16% of customers) generate ~65% of revenue.
- Purchasing is weekday-centric and time-banded (10 AM–2 PM), enabling precise communication scheduling.
- Netherlands, Ireland, and Germany are underpenetrated relative to demonstrated spend.
- Four distinct customer segments require fundamentally different engagement strategies.
- 72% repeat-purchase rate confirms strong brand loyalty; converting one-time buyers is the primary growth lever.

customer_segments.csv contains each customer's RFM scores and segment label, ready for direct CRM/marketing automation integration.

Strategic Takeaway: Protect Champions, develop Loyal Customers, win back At-Risk buyers — these three actions alone could yield a 15–25% incremental revenue uplift over 12 months.